

# Extending strong conceptual completeness via virtual ultracategories

Gabriel Saadia

Stockholm University

CT2025 – Masaryk University – Brno

# An overview

How to categorify topological spaces?

|                  |   |                  |   |                             |
|------------------|---|------------------|---|-----------------------------|
| lattice of opens | · | interior comonad | · | relational $\beta$ -modules |
| topoi            | · | ionads (Garner)  | · | virtual ultracategories     |

Relational  $\beta$ -modules = notion equivalent to topological spaces  
axiomatizing the convergence of ultrafilters instead of opens subsets.

An ultrafilter  $\mu$  converges to a point  $a$  if all neighborhoods of  $a$  are  $\mu$ -large  
we write  $\mu \succcurlyeq a$ .

For  $\mu = \delta_b$  principal, we recover the specialization order:  $\delta_b \succcurlyeq a$  iff  $b \geq a$ .

The ultraconvergence relation is an extension of the specialization order,  
that is strong enough to recover the topology.

# An overview

A topological space is compact Hausdorff iff all ultrafilters have exactly one limit point  $\rightsquigarrow$  compact Hausdorff spaces =  $\beta$ -algebras.

Lurie 2018:  $\beta$ -algebras categorify into ultracategories.

|                   |   |                 |   |                         |   |                    |
|-------------------|---|-----------------|---|-------------------------|---|--------------------|
| Compact Hausdorff | = | $\beta$ -alg    | · | rel $\beta$ -mod        | = | Topological spaces |
|                   |   | ultracategories | · | virtual ultracategories |   |                    |

Makkai 1987: conceptual completeness

A coherent topos (= coherent FO-theory)  
can be recovered from its ultracategory of points (= models).

We extend this result: a topos with enough points can be recovered  
from its virtual ultracategory of points.

1) Ultracategories

2) Virtual ultracategories

3) The proof

# The usual Stone duality

Classical propositional theories = Boolean Algebras.

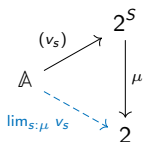
$\text{Mod}(\mathbb{A}) := \text{BoolAlg}(\mathbb{A}, 2)$  has a topology: the Stone topology.

$\mathbb{A}$  can be recovered as the clopen subsets of its space of models:

$$\mathbb{A} \cong \text{TopSp}(\text{Mod}(\mathbb{A}), 2).$$

In particular,  $\text{Mod} : \text{BoolAlg}^{\text{op}} \hookrightarrow \text{TopSp}$  is fully faithful.

The Stone topology on  $\text{Mod}(\mathbb{A})$  arises from the  $\mu \in \text{BoolAlg}(2^S, 2)$



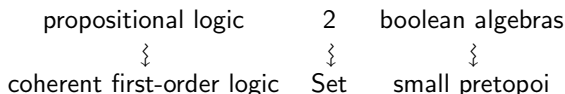
$$\begin{array}{ccc} \text{Mod}(\mathbb{A})^S & \longrightarrow & \text{Mod}(\mathbb{A}) \\ (v_s)_{s:S} & \longmapsto & \lim_{s:\mu} v_s \end{array}$$

$$\beta(S) := \text{BoolAlg}(2^S, 2) = \{\text{ultrafilters on } S\}$$

The  $\lim_{s:\mu}(-)$  determine the topology on  $\text{Mod}(\mathbb{A})$  by ultraconvergence.

# Categorifying to ultracategories

Ultracategories arise by categorifying:



We want some “categorified Stone topology” to make

$$\begin{array}{ccc} \text{Mod} : \text{Pretop}^{\text{op}} & \longrightarrow & \text{CAT} + ?? \\ \mathbb{T} & \longmapsto & \text{Pretop}(\mathbb{T}, \text{Set}) \text{ with the } ??\text{-structure} \end{array}$$

fully faithful.

$\rightsquigarrow$  the **ultrastructure** on  $\text{Mod}(\mathbb{T})$  will play the role of the categorified Stone topology.

The Stone topology on  $\text{Mod}(\mathbb{A})$  comes from  $\beta(S) = \text{BoolAlg}(2^S, 2)$

$\rightsquigarrow$  we should look at  $\text{Pretop}(\text{Set}^S, \text{Set})$ .

# Ultraproducts

Łoś's theorem: the **ultraproduct functor**

$$\begin{aligned} \int_{\mu} : \mathbf{Set}^S &\longrightarrow \mathbf{Set} \\ (A_s)_{s:S} &\longmapsto \int_{s:\mu} A_s := \operatorname{colim}_{\mu(S_0)=1} \prod_{s:S_0} A_s \end{aligned}$$

is a functor of pretopoi, i.e.,  $\int_{\mu} \in \mathbf{Pretop}(\mathbf{Set}^S, \mathbf{Set})$ .

$$\begin{array}{ccc} & \mathbf{Set}^S & \\ (M_s) \nearrow & \downarrow \int_{\mu} & \\ \mathbb{T} & & \mathbf{Set} \\ \int_{s:\mu} M_s \searrow & & \end{array}$$

induces an operation  $\int_{\mu} : \mathbf{Mod}(\mathbb{T})^S \longrightarrow \mathbf{Mod}(\mathbb{T})$ .

**ultraproducts** = “categorified ultraconverge” on  $\mathbf{Mod}(\mathbb{T})$ .

Actually, ultraproducts are enough!

$$\mathbf{Pretop}(\mathbf{Set}^S, \mathbf{Set}) \simeq \operatorname{Ind}(\mathcal{UF}_{\mathbf{Set}^S}^{\operatorname{op}}) \quad \underline{\text{Joyal 1971}}$$

i.e. a pretopos functor  $\mathbf{Set}^S \rightarrow \mathbf{Set}$  is a filtered colimit of ultraproducts.

# Ultracategories

$\text{Mod}(\mathbb{A}) \in \beta\text{-algebras} = \text{compact Hausdorff spaces}$

$\text{Mod}(\mathbb{T}) \in \beta\text{-pseudoalgebras} = \text{ultracategories}$

An **ultracategory** is a category  $\mathcal{M}$  together with:

- ultraproduct functors  $\int_{\mu} : \mathcal{M}^{\mu} \rightarrow \mathcal{M}$  ( $\mathcal{M}^{\mu} := \text{colim}_{\mu(S_0)=1} \mathcal{M}^{S_0}$ )

- functorial reindexing

$f^{\#} : \int_{s:\mu} A_s \rightarrow \int_{t:\nu} A_{f(t)}$ , for  $f : \nu \rightarrow \mu$

$$\begin{array}{ccc} \mathcal{M}^{\mu} & \xrightarrow{f^*} & \mathcal{M}^{\nu} \\ & \searrow \int_{\mu} \quad \xRightarrow{f^{\#}} \quad \swarrow \int_{\nu} & \\ & \mathcal{M} & \end{array}$$

- coherent unitor and associators  
commuting with the reindexings

$$\int_{\star} \xrightarrow{\sim} \text{Id} , \quad \int_{\sum_{s:\mu} \nu_s} \xrightarrow{\sim} \int_{s:\mu} \int_{t:\nu_s}$$

e.g.  $\text{Set}$ ,  $\text{Mod}(\mathbb{T})$ , Compact Hausdorff spaces (=  $\beta$ -algebras).



# Conceptual completeness

Makkai 1987:  $\mathbb{T}$  can be recovered from its ultracategory of models:

$\text{ev} : \mathbb{T} \xrightarrow{\sim} \text{Ult}(\text{Mod}(\mathbb{T}), \text{Set})$  is an equivalence.

$\rightsquigarrow$  conceptual completeness for coherent logic:

$\text{Mod} : \text{Pretop}^{\text{op}} \hookrightarrow \text{Ult}$  is fully faithful.

Whereas the category  $\text{Mod}(\mathbb{T}) \in \text{CAT}$  is **not enough** to recover  $\mathbb{T}$ ,  
the ultracategory of models  $\text{Mod}(\mathbb{T}) \in \text{Ult}$  is enough to recover  $\mathbb{T}$ !

How to understand this result?

**a model theoretical answer:**

ultraproducts are fundamental in the model theory of FO-logic.

**a categorical answer:**  $\text{Pretop}(\text{Set}^S, \text{Set}) \simeq \text{Ind}(\mathcal{UF}_{\text{Set}^S}^{\text{op}})$

ultraproducts are “all” the set operations that preserve FO-logic.

**a topological answer:**

ultraproducts  $\sim$  categorified ultraconvergence.

# Reconstruction theorem

Lurie 2018: extends this result to **coherent topoi**. The functor

$$\mathbf{Pretop}^{\mathrm{op}} \xrightarrow{\mathrm{Sh}} \mathbf{CohTop}$$

is faithful and conservative, but **not full**.

$$\text{coherent morphisms} \subset \text{geometric morphisms}$$

Accordingly, Lurie introduces a more flexible notion of ultrafunctors.

$$\text{ultrafunctors (Ult)} \subset \text{left-ultrafunctors (Ult}^{\mathrm{L}}\text{)}$$

Lurie's reconstruction theorem:

A coherent topos  $\mathcal{E}$  can be recovered from its ultracategory of points

$$\mathrm{ev} : \mathcal{E} \xrightarrow{\sim} \mathrm{Ult}^{\mathrm{L}}(\mathrm{pt}(\mathcal{E}), \mathbf{Set}) \text{ is an equivalence}$$

$\leadsto$  reconstruction theorem for coherent topoi:

$$\mathrm{pt} : \mathbf{CohTop} \hookrightarrow \mathrm{Ult}^{\mathrm{L}} \text{ is } \underline{\text{fully faithful}}.$$

# Towards virtual ultracategories

Lurie 2018:

$\text{ev} : \mathcal{E} \xrightarrow{\sim} \text{Ult}^{\text{L}}(\text{pt}(\mathcal{E}), \text{Set})$  is an equivalence for  $\mathcal{E}$  **coherent** topos.

He also shows:

$\text{ev} : \mathcal{E} \xrightarrow{\sim} \text{Ult}^{\text{L}}(\text{pt}(\mathcal{E}), \text{Set})$  is an equivalence for  $\mathcal{E} = \text{Sh}(\text{compact Hausdorff})$ .

None of these results contain the other!

We will generalize these both results even further.

We can recover a topos with enough points (= theory)  
from the **virtual ultracategory** structure over its points (= models).

$\rightsquigarrow$  this is a reconstruction theorem for topoi with enough points:

$\text{pt} : \text{GTop}_{\text{wep}} \hookrightarrow \text{vUlt}$  is fully faithful.

- 1) Ultracategories
- 2) Virtual ultracategories
- 3) The proof

# $\mathcal{E}$ = sheaves over a compact Hausdorff space

$X$  compact Hausdorff  $\rightsquigarrow \mathbf{pt}(X)$  the **ultracategory of its points**.

$$\begin{aligned} \mathbf{Sh}(X) &\xrightarrow{\sim} \mathbf{Ult}^{\mathbf{L}}(\mathbf{pt}(X), \mathbf{Set}) \\ E &\longmapsto \begin{cases} x \mapsto E_x \\ + \text{ coherent maps } \sigma : E_a \rightarrow \int_{x:\mu} E_x, \text{ for } a = \lim \mu. \end{cases} \end{aligned}$$

A sheaf can be reconstructed from its stalks + some data (the  $\sigma$ 's).

We generalize from  $X$  compact Hausdorff to  $T$  topological space.

We still have the construction,

$$E \in \mathbf{Sh}(T) \longmapsto \begin{cases} x \mapsto E_x \\ + \text{ coherent maps } \sigma_{a,\mu} : E_a \rightarrow \int_{x:\mu} E_x, \text{ for } a \preccurlyeq \mu. \end{cases}$$

Can we reconstruct the sheaf  $E$  from this data?

$\mathcal{E}$  = sheaves over a topological space

$$E \in \text{Sh}(T) \xrightarrow{\quad ? \quad} \left\{ \begin{array}{l} x \mapsto E_x \\ + \text{ coherent maps } \sigma_{a,\mu} : E_a \rightarrow \int_{x:\mu} E_x, \text{ for } a \preceq \mu. \end{array} \right.$$

For the reverse construction, the étale space is given by:

$$E := \bigsqcup_{x:T} E_x, \text{ opens} := \text{subsets stable by the } \sigma\text{'s.}$$

To show it is étale, we use the following characterization.

Lemma (S.) : For  $p : E \rightarrow T$  continuous,

$$p \text{ is étale} \quad \text{iff} \quad \forall \mu \succcurlyeq p(e), \exists! \nu \succcurlyeq e, p_*(\nu) = \mu \\ \text{and this } \nu \text{ is isomorphic to } \mu \text{ via } p.$$

# Categorifying ultraconvergence

$a \preccurlyeq (b_s)_{s:\mu}$  means  $\forall U \text{ open}, U \ni a \rightarrow (\forall s : \mu, U \ni b_s)$

For  $a, (b_s)_{s:\mu}$  points of  $\mathcal{E}$  topos, we denote by  $\alpha : a \multimap (b_s)_{s:\mu}$  for

$$\begin{array}{ccc}
 & & \text{Set}^S \\
 & \nearrow^{(b_s^*)} & \downarrow \int_\mu \\
 \mathcal{E} & & \text{Set} \\
 & \searrow_{a^*} & \\
 & & 
 \end{array}
 \quad \uparrow \alpha \quad
 \alpha \in \text{Nat}_{E:\mathcal{E}}(E_a, \int_{s:\mu} E_{b_s})$$

it is a **proof-relevant** version of “ $a \preccurlyeq (b_s)_{s:\mu}$ ”.

$a \multimap (b_s)_{s:\mu}$  is an **ultraarrow** with codomain an ultrafamily of objects, these ultraarrows form a **generalized multicategorical structure** on the points of  $\mathcal{E}$  that we denote  $\text{pt}(\mathcal{E}) \in \mathbf{vUlt}$ .

$\rightsquigarrow$  the notion of **virtual ultracategory**.

(generalized multicategories are introduced in Cruttwell and Shulman 2010)

# Virtual ultracategories

A **virtual ultracategory** is a generalized multicategory with arrows having the shape  $a \multimap (b_s)_{s:\mu}$ . Explicitly:

- a class of objects  $X$ ,
- a functor  $\text{Hom} : X \times \beta(X) \rightarrow \text{Set}$  (+ identity, composition...)

where  $\beta(X) := \mathcal{U}\mathcal{T}_{\text{Set}^X}^{\text{op}}$  is the category of ultrafamilies in  $X$ .

(in particular  $\text{id}_a : a \multimap (a)_*$  induces an arrow  $a \multimap (a)_\mu$ )

The **virtual ultracategory of points** of a topos is defined as above

$$\text{pt} : \text{GTop} \longrightarrow \text{vUlt}$$

with  $\text{Ob}(\text{pt}(\mathcal{E})) := \text{pt}(\mathcal{E})$  and  $\text{Hom}(a, (b_s)_{s:\mu}) := \text{Nat}_{E:\mathcal{E}}(E_a, \int_{s:\mu} E_{b_s})$ .

Ultracategories are recovered as the **representable** virtual ultracategories.



# Statement of the theorem

Theorem (S.): Let  $\mathcal{E}$  be a topos with enough points, the functor

$\text{ev} : \mathcal{E} \xrightarrow{\sim} \mathbf{vUlt}(\mathbf{pt}(\mathcal{E}), \mathbf{Set})$  is an equivalence.

Some particular cases:

- for  $\mathcal{E}$  **coherent**:  
amounts to Lurie's reconstruction theorem.
- for  $\mathcal{E} = \mathbf{Sh}(\text{topological space})$ :  
amounts to the above lemma characterizing étale maps.
- for  $\mathcal{E}$  **preasheaf topos**:  
follows from Yoneda.

This theorem generalizes, and gives a new proof, to Lurie's result.

- 1) Ultracategories
- 2) Virtual ultracategories
- 3) The proof

# Strategy

We already proved the case  $\mathcal{E} = \text{Sh}(\text{topological space})$ ; to extend from the topological case to the general case, we use **representation of topoi by topological groupoids**.

Butz and Moerdijk 1998: any topos with enough points  $\mathcal{E}$  can be represented by a **topological groupoid**  $(T_\bullet)$ ,

i.e. there is a **universal descent cocone**:

$$\text{Sh}(T_1 \times_{T_0} T_1) \xrightarrow{-m \rightarrow} \text{Sh}(T_1) \begin{array}{c} \xrightarrow{s} \\ \xleftarrow{u} \\ \xrightarrow{t} \end{array} \text{Sh}(T_0) \quad \begin{array}{c} \xrightarrow{\pi} \\ \searrow \end{array} \mathcal{E}$$

and so,  $\mathcal{E} = \text{Sh}_{\text{eq}}(T_\bullet)$  the topos of equivariant sheaves over  $(T_\bullet)$ .

# A problem!

$$\mathrm{Sh}(T_1 \times_{T_0} T_1) \longrightarrow \mathrm{Sh}(T_1) \begin{matrix} \xrightarrow{\quad} \\ \xleftarrow{\quad} \end{matrix} \mathrm{Sh}(T_0) \begin{matrix} \xrightarrow{\quad \pi \quad} \\ \searrow \mathcal{E} \end{matrix} \quad \text{GTop}$$

$$\mathrm{pt}(T_1) \times_{\mathrm{pt}(T_0)} \mathrm{pt}(T_1) \longrightarrow \mathrm{pt}(T_1) \begin{matrix} \xrightarrow{\quad} \\ \xleftarrow{\quad} \end{matrix} \mathrm{pt}(T_0) \begin{matrix} \xrightarrow{\quad \mathrm{pt}(\pi) \quad} \\ \searrow \mathrm{pt}(\mathcal{E})|_X \end{matrix} \quad \text{vUlt}$$

( $X$  is the image of the points of  $T_0$  by  $\pi$ )

$$\begin{aligned} \mathcal{E} &\simeq \mathrm{Cocone}_{\mathrm{desc}}(\mathrm{Sh}(T_\bullet); \mathrm{Set}[0]) \simeq \mathrm{Cocone}_{\mathrm{desc}}(\mathrm{pt}(T_\bullet); \mathrm{Set}) \stackrel{??}{\simeq} \mathrm{vUlt}(\mathrm{pt}(\mathcal{E})|_X, \mathrm{Set}) \\ &\simeq \mathrm{Sh}_{\mathrm{eq}}(T_\bullet) \qquad \qquad \qquad \simeq \mathrm{Sh}_{\mathrm{eq}}(\mathrm{pt}(T_\bullet)) \end{aligned}$$

$\mathcal{E}$  is the colimit of  $(\mathrm{Sh}(T_\bullet))$ , we want  $\mathrm{pt}(\mathcal{E}|_X)$  to be colimit of  $(\mathrm{pt}(T_\bullet))$ .

i.e. we want  $\mathrm{pt}(\pi)$  to be effective descent.

Problem:  $\mathrm{pt} : \mathrm{GTop} \rightarrow \mathrm{vUlt}$  does not preserve colimits (right adjoint!)

# A solution: ample indexings

Solution: it works for a well-chosen groupoid!

- 1) We prove a proposition ensuring a functor of virtual ultracategories to be effective descent.

Proposition: A functor of v-ultracats  $F : X \rightarrow Y$  surjective on objects and such that any  $F(x) \multimap (b_s)_{s:\mu}$  can be lift to some  $x \multimap (y_s)_{s:\mu}$  is effective descent.

- 2) We consider the groupoid  $(T_{\bullet}^{\text{amp}})$  of small models with an **ample** indexing, so that  $\text{pt}(\pi)$  satisfies the property of 1.

Definition: An indexing  $\alpha : \kappa \rightrightarrows M$  is ample if  $|\kappa \setminus \text{dom}(\alpha)| = |\kappa|$ .

$$\begin{array}{ccc} \kappa & & \\ \alpha \downarrow & \searrow (\beta_s) & \\ M_p & \longrightarrow \int_{s:\mu} M_{q_s} & \end{array}$$

$\alpha$  ample  $\leadsto$  we can extend to surjective  $(\beta_s)$ .

- 3) A theorem of **Wrigley 2023** ensures that the ample condition is not too strong, i.e., that  $(T_{\bullet}^{\text{amp}})$  represents  $\mathcal{E}$ .

# The duality

We have shown:

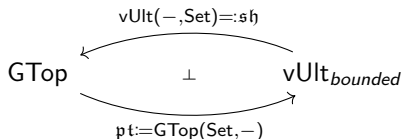
$ev : \mathcal{E} \xrightarrow{\sim} \mathbf{vUlt}(\mathbf{pt}(\mathcal{E})_{\upharpoonright X}, \mathbf{Set})$  is an equivalence for  $X \subseteq \mathbf{pt}(\mathcal{E})$  small separating.

We can generalize:

$ev : \mathcal{E} \xrightarrow{\sim} \mathbf{vUlt}(\mathbf{pt}(\mathcal{E}), \mathbf{Set})$  is an equivalence for any  $\mathcal{E}$  with enough points.

Proof: note that  $\mathbf{pt}(\mathcal{E}) \simeq \operatorname{colim}_A \mathbf{pt}(\mathcal{E})_{\upharpoonright A}$ , where  $A$  range over small separating sets of points of  $\mathcal{E}$ , to reduce to the above case.  $\square$

We get a **pseudoidempotent 2-adjunction**



inducing a fully faithful embedding  $\mathbf{pt} : \mathbf{GTop}_{\text{wep}} \hookrightarrow \mathbf{vUlt}$ .

# Summary

We introduced the notion of **virtual ultracategories** that categorifies relational  $\beta$ -modules.

We categorified the equivalence between topological spaces and relational  $\beta$ -modules to a **pseudoidempotent 2-adjunction**.

$$\begin{array}{ccc} & \text{vUlt}(-, \text{Set}) =: \mathfrak{sh} & \\ \text{GTop} & \xleftarrow{\quad} & \text{vUlt}_{\text{bounded}} \\ & \text{pt} := \text{GTop}(\text{Set}, -) & \end{array} \quad \perp$$

This adjunction induces a **fully faithful embedding**  $\text{pt} : \text{GTop}_{\text{wep}} \hookrightarrow \text{vUlt}$  extending Lurie's reconstruction theorem to all topoi with enough points.

Virtual ultracategories fit in the framework of generalized multicategories of Cruttwell and Shulman 2010, and ultracategories are recovered as the **representable** ones.

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## Proof of the 0-dimensional case

We prove the non-trivial implication,

$$p \text{ is étale} \iff \forall \mu \succcurlyeq p(e), \exists! \nu \succcurlyeq e, p_*(\nu) = \mu$$

and this  $\nu$  is isomorphic to  $\mu$  via  $p$ .

- $p_*$  is étale
- the hypothesis gives a lift  $\sigma$
- the topological fact  $(\star)$  gives  $V$
- we pullback along  $\delta$   
not continuous!
- $\delta^* T \subseteq T$  is not open  
 and  $\xi$  not continuous:  
 we restrict to  $W \subset V$ .

$$W := \{w \in \delta^* V \mid \forall \mu \succcurlyeq w \text{ } (\xi \text{ is defined on } \mu \text{ and } \xi_*(\mu) \succcurlyeq \xi(w))\}$$

(★) a section on a compact of a Hausdorff space can be extended to an open.  $\square$

